MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 5AWP2

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus, flat face, single beam, multiband post accelerator

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ec1	Eb1	Eb2	Eb3	ed	Rg	Zd	Ehk	Eb3/Eb2	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	v	Meg	Meg	Vdc	Ratio	ft
Maximum:	6.9	0	1,700	3,850	16,500 (Note 1)	1,320	1.5	1.0 (Note 2)	±180	4.5 (Note 3)	10,000
Minimum:	5.7	-200	---	1,000	4,500	---	---	---	---	---	---
TEST CONDITIONS:	6.3	Adjust	Focus	2,300	10,300	---	---	1.0	---	4.5	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
5201	Voltage breakdown		0.65	II	---	---	---	---
5201	Voltage breakdown (electrostatic types)		0.65	II	---	---	---	---
5206	Gas "cross"	Ib3 = 50 µAdc (see note 9)	0.65	II	---	---	---	---
5108	Screen and faceplate blemishes		0.65	II	---	---	---	---
5223	Modulation	Ib3 = 25 µAdc	0.65	II	ΔEc1	---	35	Vdc
5231	Spot position (electrostatic deflection)		0.65	II	---	---	15	mm
5231	Spot displacement (leakage)		0.65	II	Displ	---	10	mm
5241	Grid cutoff voltage		0.65	II	Ec1	-35	-85	Vdc
---	Pattern distortion	See note 7	0.65	II	---	---	---	---
5251	Grid No. 1 leakage current		0.65	II	Ic1	---	3	µAdc
5251	Anode No. 1 leakage current		0.65	II	Ib1	---	5	µAdc
5251	Anode No. 2 leakage current		0.65	II	Ib2	---	5	µAdc
	Quality conformance inspection, part 3							
1301	Heater current		6.5	S3	If	540	660	mA
5201	Electrode current (anode No. 1)		6.5	S3	Ib1	-15	10	µAdc
(D) 5201	Electrode current (cathode)	Ib3 = 50 µAdc (see note 4)	6.5	S3	Ik	---	1,000	µAdc
5101	Base alinement (electrostatic types)	+ 3D4, pin No. 5	6.5	S3	---	---	---	---
5101	Side terminal alinement (electrostatic types)	+ 3D4	6.5	S3	---	---	---	---
5101	Side terminal and base alinement (electrostatic types)	Pin No. 5	6.5	S3	---	---	---	---
5101	Neck and base alinement (electrostatic types)		6.5	S3	---	---	---	---
(D) 5101	Angle between traces		6.5	S3	---	89	91	Degrees
(D) 5216	Stray light emission (conventional types)	Eb2 = 3,850 Vdc; Eb3 = 16,500 Vdc	6.5	S3	---	---	---	---
5221	Screens		6.5	S3	cb1	370	---	cb
(D) 5226	Line width "A" (electrostatic deflection)	Ib3 = 25 µAdc	6.5	S3	Width	---	0.40	mm

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 2 -Continued							
5226	Line width "B" (electrostatic deflection)	Ib3 = 25 µAdc	6.5	S3	Width	---	0.60	mm
Ⓓ5246	Focusing voltage at cutoff		6.5	S3	Eb1	---	730	Vdc
5246	Focusing voltage at modulation condition		6.5	S3	Eb1	500	---	Vdc
Ⓓ5248	Deflection factor (1D2)	See note 5	6.5	S3	DF	113	128	Vdc/in.
Ⓓ5248	Deflection factor (3D4)		6.5	S3	DF	113	128	Vdc/in.
5251	Heater-cathode leakage		6.5	S3	Ihk	---	10	µAdc
1101	Secureness of base, cap, or insert		6.5	S3	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3					
---	Life-test provisions	Group C; Ib3 = 12.5 µAdc; t = 500 hours (min)	---	---	---	---
---	Life-test end points	Ib3 = 18.75 µAdc, modulation	ΔEc1	---	25	Vdc
		Line width "A"	Width	---	0.40	mm
		Line width "B"	Width	---	0.60	mm
		Heater-cathode leakage	Ihk	---	30	µAdc
		Grid No. 1 leakage	Ic1	---	5	µAdc
		Anode No. 1 leakage	Ib1	---	15	µAdc
		Anode No. 2 leakage	Ib2	---	15	µAdc
Ⓓ	Periodic-check tests					
5101	Neck and bulb alinement (electrostatic deflection)		Dia	---	2.25	Inch
5101	Face tilt		---	---	---	---
5216	Cathode illumination		---	---	10×10^{-6}	fL
5248	Deflection-factor uniformity		---	---	2	%
---	Deflection defocusing	See note 6	---	---	1:2	Ratio
5256	Magnetization		---	---	---	---
1331	Direct-interelectrode capacitance	See note 8				
		g1 to all	Cg1	---	5.0	pF
		k to all	Ck	---	4.1	pF
		D1 to D2	C1D2	---	2.2	pF
		D3 to D4	C3D4	---	1.5	pF
		D1 to all	CD1	---	5.4	pF
		D2 to all	CD2	---	6.1	pF
		D3 to all	CD3	---	4.5	pF
		D4 to all	CD4	---	3.9	pF

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METHOD	REQUIREMENT OR TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
				MIN	MAX	
Ⓓ	Periodic-check tests -Continued					
1141	Pressure (implosion)		---	---	---	---
5111	Vibration		Width	---	1.0	mm

NOTES:

1. Accelerator to final intensifier electrode voltage equally divided over the three intensifier electrodes by 25 megohm resistors connected between each intensifier and the accelerator.
2. It is recommended that the deflection electrode circuit resistances be 1 megohm, or less, and approximately equal, otherwise beam shift at high drives can be expected. Higher resistance values up to 5 megohms may be used for low-beam current operation.
3. This tube is designed for optimum performance when operating at an Eb3/Eb2 ratio of 4.5. Operation at other ratios of Eb3/Eb2 may result in changes in deflection uniformity and pattern distortion.
4. Accelerator power input (average) shall be limited to 6 watts.
5. 1D2 deflection factor shall be within 8 volts of 3D4. The useful scan in the 1D2 and 3D4 directions is plus or minus 2 (50.8 mm) inches from tube face center.
6. The ratio of the line width of a 4 (101.6 mm) - inch long trace, adjusted for best center focus, to the line width measured on the faceplate axis when the trace is deflected perpendicularly plus or minus 2 (50.8 mm) inches from the face center by a balanced dc voltage, shall not be greater than the value specified.
7. All portions of a raster pattern, adjusted so its widest points just touch the sides of a 3.040 (77.22 mm) - inch square, will fall within the area bounded by the 3.040 (77.22 mm) - inch square and an inscribed 2.960 (75.18 mm) - inch square.
8. All other electrodes contained within the envelope shall be tied together.
9. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

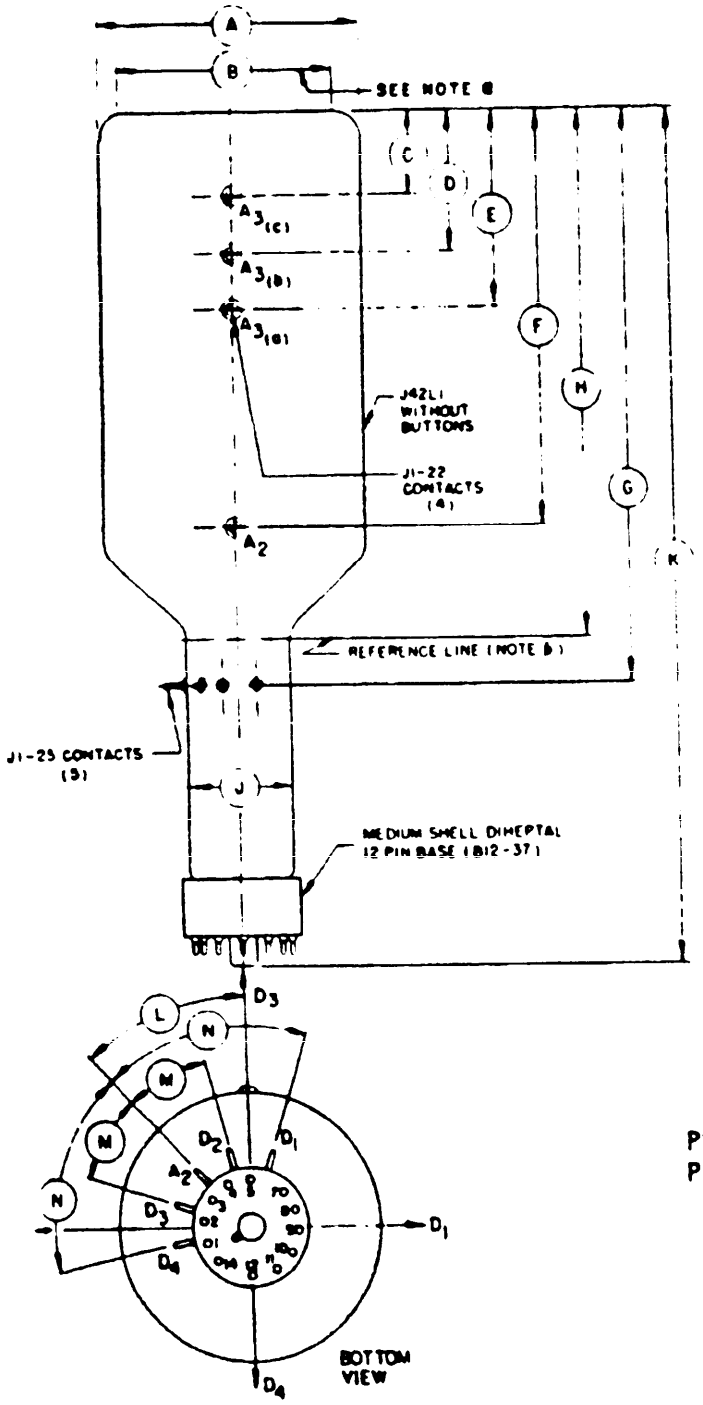
Review activities:
Army - EL
Navy -
Air Force - 11, 85
DSA - ES

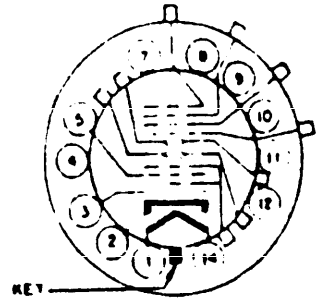
User activities:
Army - ME
Navy - AS, OS, MC, CG, SH
Air Force - 19

Preparing activity:
Navy - EC

(Project 5960-2358)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	5.156 (130.96)	5.344 (135.74)
B	4.500 (114.30)	
C	1.250 (31.75)	1.625 (41.28)
D	2.375 (60.33)	2.750 (69.85)
E	3.500 (88.90)	3.875 (98.43)
F	7.812 (198.42)	8.812 (223.82)
G	10.750 (273.05)	11.250 (285.75)
H	10.062 (255.57)	10.437 (265.10)
J	1.937 (49.20)	2.062 (52.37)
K	16.500 (419.10)	17.000 (431.80)
L	35°	55°
M	25°	35°
N	55°	65°

PIN CONNECTIONS

Pin No.	Element
1	Heater
2	Cathode
3	Grid No. 1
4	Internal connection
5	Focusing electrode
7–12	No connection
14	Heater

NOTES:
a. Minimum useful screen diameter.
b. Reference line, point where ring gage 2.063 +.0011 (52.40 +.03 mm) ID, .500 (12.70 mm) long will stop.

FIGURE 1. Outline drawing of electron tube type 5AWP2.

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